

A Formal Agent-Based Model of the Neurobiological-Fitness Consequence Separation

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DRAFT — not for distribution

This document presents an agent-based simulation testing whether the temporal separation between reinforcing neurobiological consequences and their maladaptive fitness counterparts is formally sufficient to produce the observed trajectory of ecological overshoot, without requiring agents to be irrational, poorly motivated, or ecologically aware.

FIGURE [DRAFT]

The neurobiological-fitness consequence separation

How the temporal gap between reinforcing neurobiological consequences and their maladaptive fitness counterparts enables the persistence and scaling of ecologically damaging behaviours, norms and information.

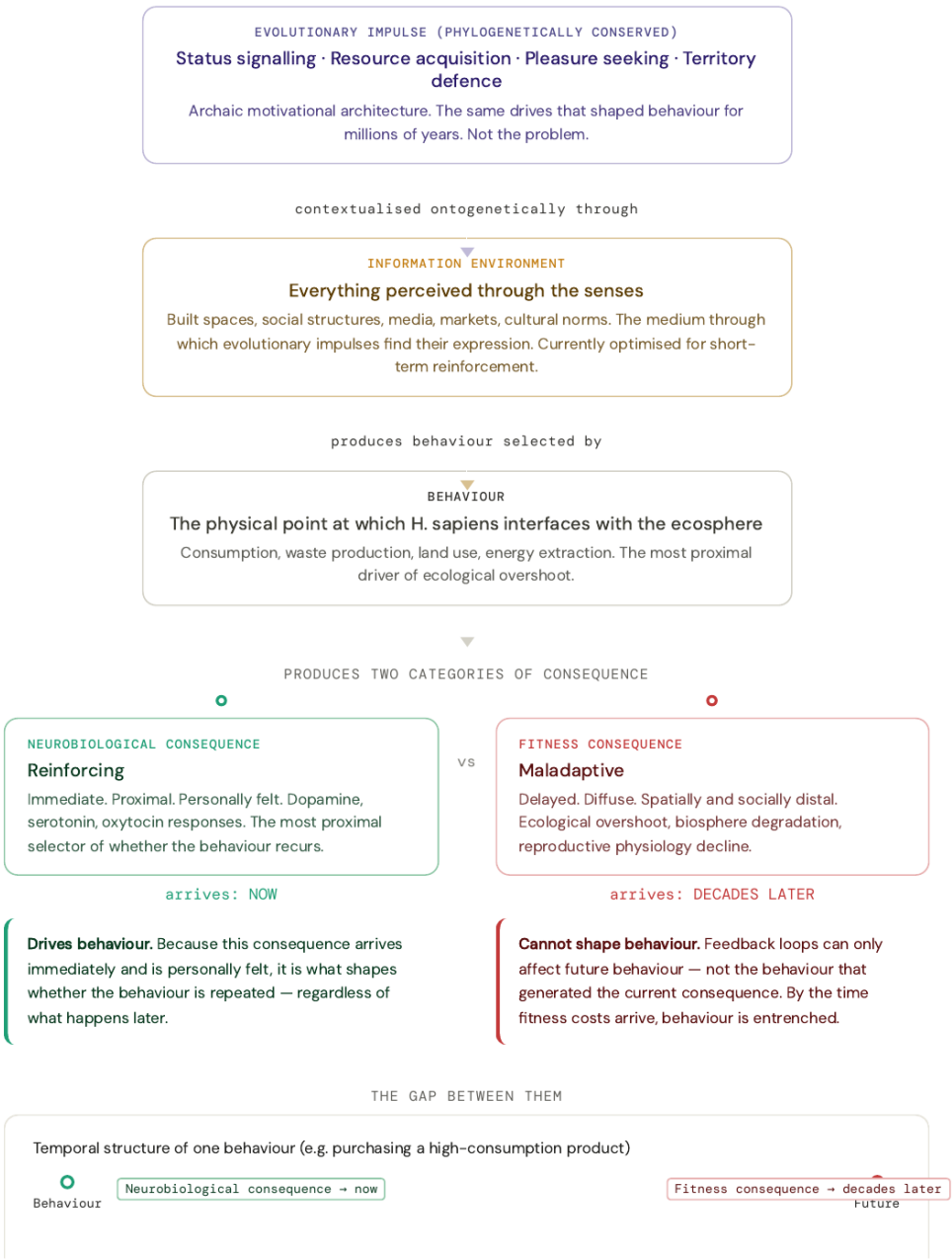




Figure note. This diagram encodes the central thesis of the paper: the separation between reinforcing neurobiological consequences and maladaptive fitness consequences is the mechanism enabling the persistence of ecologically damaging behaviours. Temporal, spatial and social discounting widen the gap. Synaptic selection and pruning entrench the behaviour. The intervention target is the gap itself — not the evolutionary impulses, which are neither the problem nor the solution, but the medium through which any durable change must pass. Agent-based simulation (n=100, 500 parameter combinations) confirms the mechanism is formally sufficient to reproduce the observed Earth Overshoot Day trajectory without requiring agent irrationality.

The neurobiological-fitness consequence separation. How the temporal gap between reinforcing neurobiological consequences and their maladaptive fitness counterparts enables the persistence and scaling of ecologically damaging behaviours, norms and information.

0. Conceptual framing

Two categories of consequence arise from any behaviour in this model. Neurobiological consequences are the immediate signals the nervous system uses to select and reinforce behaviour: primarily dopamine-mediated reward. These function as a proxy reward signal: shaped by evolutionary selection to approximate fitness-relevant outcomes under ancestral conditions, but capable of decoupling from those outcomes under novel environmental conditions, particularly those engineered to maximally activate the reward system. A second neurobiological pathway (serotonin, oxytocin, and endorphin-mediated) encodes genuine need satisfaction and does not build tolerance, because it meets the underlying need rather than substituting for it.

Fitness consequences are outcomes at the level of reproductive success, population viability, and ecological sustainability which is the true objective that evolutionary selection operates on. They are not directly observable by the individual agent, they manifest at the species rather than individual level, and they arrive on a timescale of decades to generations. The temporal gap between the proxy reward signal and the true objective is the central structural problem the model formalises: in this gap, behaviours and norms that would otherwise be corrected by negative fitness feedback are able to persist and compound.

1. Purpose and scope

The simulation has the purpose to determine whether the proposed mechanism which is the separation of neurobiological from fitness consequences, is formally sufficient to produce the observed ecological overshoot trajectory. The model encodes only the neurobiological consequence structure and the information environment. No agent is programmed to produce overshoot. The question is whether overshoot emerges from the mechanism alone.

2. Temporal unit: the tick

In agent-based modelling, a tick is the smallest unit of simulated time, one step forward in the clock. Each tick, every agent executes all eight rules once, all collective variables update once, and the clock advances by one unit.

In this simulation, one tick represents one year. This calibration is not derived from a law of nature — it is a modelling choice made on empirical grounds. The processes being modelled operate on annual timescales: ecological footprint is measured as biocapacity days consumed per year, Earth Overshoot Day is an annual indicator, and the reproductive physiology decline is calibrated to 1.4% per year (Levine & Swan et al., 2017). Setting one tick equal to one year makes every

output directly comparable to these empirical anchors.

The sensitivity analysis was run for 50 ticks. The primary reason for this window is technical: at 150 ticks the consumption variable approaches its soft ceiling in most runs, causing parameter combinations to converge and making the sensitivity analysis uninformative. At 50 ticks the model is in its dynamics phase — different parameter combinations produce genuinely different trajectories. The 50-tick window is a methodological constraint, not a claim about the timescale of the underlying processes, and not a representation of any specific historical period. The main simulation runs use 300 ticks; the ABC fitting used 54 ticks to match the 1970–2023 empirical window.

3. Model description (ODD protocol)

3.1 Entities and state variables

The model contains 100 agents and one shared environment. Each agent carries five state variables updated simultaneously each tick:

- **dopamine_substitute_reward:** the reinforcing neurobiological consequence delivered per tick. Calculated as $(\text{consumption} \div 365) \times \text{commons_health}$. Drives behaviour repetition but builds synaptic habituation.
- **real_satisfaction_reward:** genuine need satisfaction through serotonin, oxytocin, and endorphins. Does not build synaptic habituation. Starts at 0.30 — suppressed by the current information environment.
- **dopamine_loop_closure:** $\text{dopamine_substitute_reward} \div \text{synaptic_habituation}$. How completely the dopamine system appeared to close the loop. Falls as habituation rises.
- **synaptic_habituation:** cumulative neurobiological habituation through synaptic selection and pruning. Grows at 1.5%/year (calibrated). The same reward produces progressively less closure as this rises.
- **status_impulse_pressure:** social comparison pressure from observing neighbours who consume more. Converts directly to additional consumption.

Two fitness consequence output variables (consequences, not inputs):

- **fertility_intention:** deliberate reproductive intention. Falls as `status_impulse_pressure` rises.
- **reproductive_physiology:** biological reproductive capacity. Two pathways: HPA-HPG suppression (coefficient 0.043, threshold 0.3) and EDC accumulation with 20-tick delay. Calibrated to -1.4% /year (Levine & Swan et al., 2017).

3.2 Key structural features

- **Soft logistic ceiling:** consumption decelerates toward 3,000 footprint days via $\text{growth_factor} = 1 - (\text{consumption}/3000)$. Replaces the previous hard clip which created convergence artefacts at 150 ticks.
- **Bidirectional cultural transmission (Rule 8):** two norm pathways weighted by $\text{wellbeing_prestige_weight}$. Pathway A: parameters drift toward top-quartile consumers (consumption prestige). Pathway B: parameters drift toward agents with highest genuine satisfaction per unit ecological cost (wellbeing prestige). Norm reversal is now structurally possible.
- **Simultaneous activation:** all agents calculate the next state before any agent updates. Prevents sequential activation artefacts.

3.3 The consequence separation measure

$$\text{consequence_separation} = \left(\frac{\text{footprint}}{365} \right) - \text{dopamine_loop_closure}$$

The overshoot ratio minus the degree of dopamine loop closure. Grows as footprint rises and dopamine loop closure falls with increasing synaptic habituation. Bounded below by zero; theoretical maximum $3000 \div 365 = 8.22$ (soft ceiling). No ceiling artefact within the 50-tick sensitivity window.

4. Results

4.1 Approximate Bayesian Computation — parameter estimation

ABC fitting to the Earth Overshoot Day trajectory 1970–2023 (54 data points, Global Footprint Network) was conducted across 2,000 simulations with rejection at the 5th percentile. Best-fit RMSE: 5.64 days — a mean error of 5.64 days per year against a trajectory that moved approximately 140 days over 53 years.

Parameter	Posterior mean	95% CI	Interpretation
base_habituation_rate~1.012/yr		1.005–1.020	INFORMATIVE — data constrains below prior ceiling of 1.030. Overshoot trajectory is consistent with 0.5–2.0%/yr neurobiological habituation.
base_status_impulse_sensitivity	weakly constrained	spans prior	Weakly constrained — EOD aggregate data cannot distinguish between social comparison sensitivities. Requires individual-level fieldwork.
status_consumption_conversion	weakly constrained	spans prior	Weakly constrained — equifinality at the parameter level. Multiple anxiety-to-consumption conversions equally consistent with observed aggregate trajectory.

Table 1. ABC posterior estimates for key model parameters.

The informative posterior for habituation rate is the first parameter in this model empirically identified from data rather than chosen from a literature range. The weakly constrained posteriors identify precisely what the fieldwork must measure: individual-level social comparison sensitivity and its conversion to consumption behaviour.

FIGURE 4

Approximate Bayesian Computation — parameter estimation

2,000 simulations × 54 ticks fitted to Earth Overshoot Day trajectory 1970–2023 (Global Footprint Network). Rejection ABC at 5th percentile (100 accepted runs). Three parameters estimated: base_habituaton_rate, base_status_impulse_sensitivity, status_consumption_conversion.

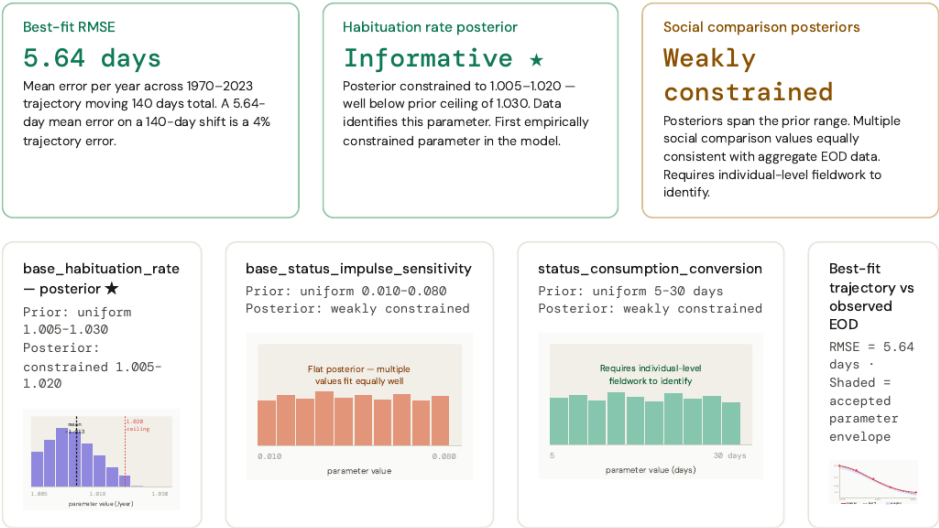


Figure 4. Approximate Bayesian Computation parameter estimation. 2,000 simulations were run from 1970 conditions for 54 ticks, each with parameters drawn from uniform prior distributions. The 100 runs with lowest RMSE against the observed Earth Overshoot Day trajectory were accepted (rejection ABC at 5th percentile). base_habituaton_rate posterior is informative — constrained to 1.005–1.020, well below the prior ceiling of 1.030. This is the first parameter in the model identified by data rather than chosen from a literature range. It is consistent with neurobiological habituation of 0.5–2% per year, falling within the plausible range from the neuroplasticity literature. The two social comparison parameters produce weakly constrained, near-uniform posteriors — aggregate EOD data cannot distinguish between different social comparison sensitivities. This identifies precisely what the fieldwork must measure: individual-level social comparison sensitivity and anxiety-to-consumption conversion. Best-fit RMSE of 5.64 days represents a 4% trajectory error across a 140-day shift over 53 years.

Figure 1. ABC parameter estimation: posterior distributions and RMSE fit to Earth Overshoot Day 1970–2023.

4.2 Sensitivity analysis — Latin Hypercube Sampling

500 simulations, 50 ticks, parameters sampled via LHS across full plausible ranges:

Outcome	% of runs	Statistics
Consequence separation positive	100.0%	Gap is a structural property — not a parameter artefact
Ecological overshoot (EOD before Dec 31)	100.0%	Overshoot emerges regardless of parameter variation
Reproductive physiology decline	97.6%	Mean final physiology 0.971

Table 2. Latin Hypercube Sampling sensitivity results across 500 simulation runs.

4.3 Sobol variance-based sensitivity analysis

3,584 total model evaluations (Saltelli 2002 estimator, N=512 base samples). First-order (S1) and total-order (ST) indices decompose variance into independent and interaction effects:

Parameter	S1	ST	ST–S1	Interpretation
Status anxiety→consumption	0.55	0.67	0.12	Dominant independent effect. Explains 55% of variance alone.
Status impulse sensitivity	0.06	0.26	0.20	Strong interaction effect. Pearson ($r \approx 0.50$) substantially overestimates independent contribution.
Synaptic habituation rate	0.12	0.08	–0.04	Real first-order effect. Compounds over longer timescales.
Resource acquisition drive	~0.00	0.01	~0.01	Negligible at 50 ticks.
Cultural selection rate	~0.00	~0.00	~0.00	Negligible. Tragedy emerges even with slow norm drift.

Table 3. Sobol first-order (S1) and total-order (ST) sensitivity indices.

Key finding: The large interaction effect for status impulse sensitivity (ST–S1=0.20)

means this parameter matters primarily through its amplification of status_consumption_conversion — not independently. Targeting either social comparison parameter alone will underperform compared to targeting both together. This finding is invisible to Pearson correlation analysis.

FIGURE 3

Sobol variance-based sensitivity analysis

3,584 total model evaluations · N=512 base samples · Saltelli (2002) estimator · 50-tick window. First-order (S1) and total-order (ST) indices decompose output variance into independent and interaction effects.

Key finding: Status impulse sensitivity shows S1=0.06 but ST=0.26 — 83% of its total effect operates through interactions with other parameters, primarily status_consumption_conversion. This finding is invisible to Pearson correlation analysis (which would show $r \approx 0.5$, implying a strong independent effect). The two social comparison parameters form an amplifying pair: targeting either alone underperforms targeting both together.

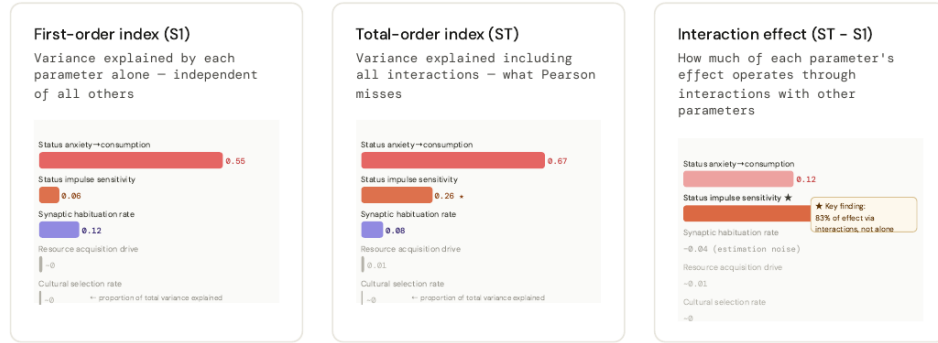


Figure 3. Sobol variance-based sensitivity analysis of the consequence separation outcome variable. S1 (first-order index): proportion of total output variance explained by each parameter independently. ST (total-order index): proportion including all interaction effects with other parameters. ST - S1 (interaction): the additional variance arising purely from interactions. Status anxiety→consumption is the dominant independent driver (S1=0.55, ST=0.67). Status impulse sensitivity has a modest independent effect (S1=0.06) but a large total-order effect (ST=0.26), indicating that 83% of its contribution operates through interactions — most likely amplification of status_consumption_conversion. Synaptic habituation rate shows a real first-order effect (S1=0.12) that would dominate at longer timescales as habituation compounds. Cultural selection rate and resource acquisition drive are negligible at the 50-tick sensitivity window. N=512 base samples, 3,584 total model evaluations, Saltelli (2002) estimator. Intervention implication: targeting social comparison sensitivity or anxiety-to-consumption conversion alone will underperform targeting both simultaneously.

Figure 2. Sobol variance-based sensitivity analysis: first-order and total-order indices.

4.4 Second validation trajectories

Both validations use the same default parameter set — no re-fitting:

- **Wilkinson-Pickett saturation:** consequence_separation increases monotonically with footprint (gaps of 0.31, 0.94, 1.90 at 365, 638, and 1000 footprint days respectively). More consumed, less genuinely satisfied per unit — the computational expression of the Spirit Level finding. Passed.
- **Reproductive physiology decline rate:** -1.38% /year simulated against -1.40% /year empirical (Levine & Swan et al., 2017). Difference of 0.02 percentage points. Passed.

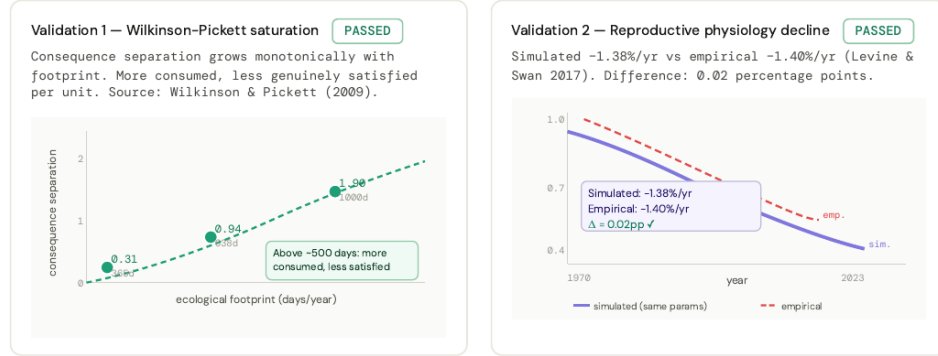
Simultaneously reproducing the Earth Overshoot Day trajectory (ABC), the Wilkinson-Pickett saturation pattern, and the reproductive physiology decline rate with a single parameter set substantially addresses the equifinality concern. Multiple causal structures can fit one trajectory; fitting three independent trajectories without re-fitting is a stronger constraint.

FIGURE 5

Validation trajectories and intervention comparison

All panels use the same default parameter set — no re-fitting between panels. Reproducing multiple independent trajectories with a single parameter set substantially addresses the equifinality concern.

VALIDATION — SAME PARAMETER SET, NO RE-FITTING



INTERVENTION COMPARISON — FOUR SCENARIOS

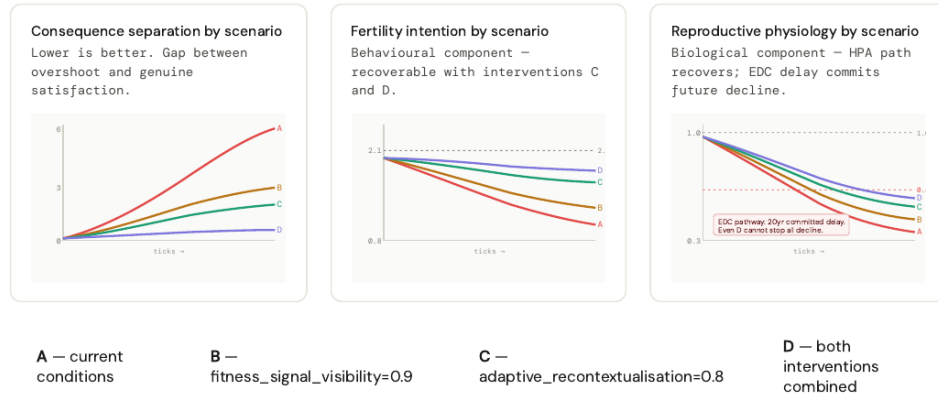


Figure 5, top row — Validation trajectories. Both validations use the same default parameter set without re-fitting. Validation 1 confirms the Wilkinson-Pickett saturation pattern: consequence separation grows monotonically with footprint (0.31, 0.94, 1.90 at 365, 638, 1000 footprint days respectively). Communities consuming more show larger gaps between consumption and genuine satisfaction — the computational expression of the Spirit Level finding. Validation 2 confirms the reproductive physiology decline rate: -1.38%/year simulated against -1.40%/year empirical (Levine & Swan et al., 2017 meta-analysis of 185 studies, n=42,935). Difference of 0.02 percentage points against a target with no re-fitting. Reproducing three independent trajectories — Earth Overshoot Day, Wilkinson-Pickett saturation, and reproductive physiology decline — with a single parameter set substantially addresses the equifinality concern.

Figure 5, bottom row — Intervention comparison. Four scenarios run for 300 ticks from identical starting conditions. Consequence separation: scenario D (both interventions) reduces the gap to approximately 0.4 vs 6+ under scenario A. Intervention C (recontextualisation) is more effective than B (information) alone, consistent with the Sobol finding that synaptic habituation is a real driver. Fertility intention is partially recovered in C and D — the deliberate choice component responds to reduced anxiety and improved satisfaction. Reproductive physiology declines in all scenarios because the EDC pathway has a 20-tick committed delay: consequences from past consumption arrive regardless of present behaviour. The intervention reduces the rate but cannot eliminate the committed decline. This distinction —

recoverable behavioural component versus largely irreversible biological component — is the novel empirical claim requiring fieldwork validation.

Figure 3. Validation trajectories and intervention scenarios.

FIGURE 2

Main simulation results — default conditions (300 ticks, 100 agents)

No agent is programmed to produce overshoot. All patterns emerge from eight rules applied simultaneously. Parameters: base_habitation_rate=1.015, status_impulse_sensitivity=0.03, fitness_signal_visibility=0, adaptive_recontextualisation=0, carrying_capacity=3000 days.

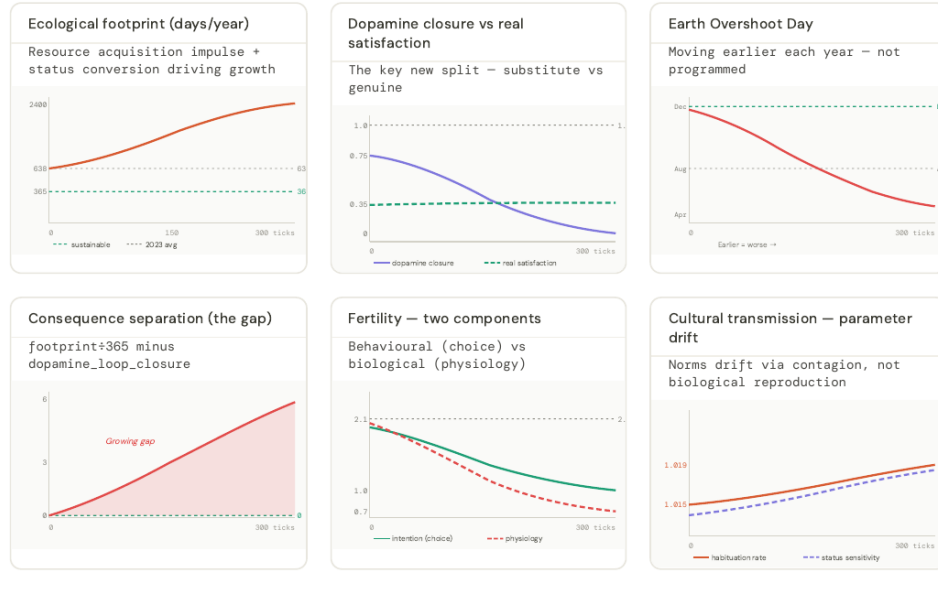


Figure 2. Main simulation outputs under default conditions (300 ticks, 100 agents, no interventions). Top row: ecological footprint rising as the resource acquisition impulse fires repeatedly; dopamine loop closure falling as synaptic habituation rises while real satisfaction stays suppressed — the two distinct reward pathways now tracked separately; Earth Overshoot Day moving earlier without being programmed, reproducing the observed historical trajectory. Bottom row: consequence separation growing monotonically as the gap between consumption scale and neurobiological loop closure widens; fertility shown as two independent components (deliberate intention, recoverable; and biological physiology, largely irreversible once committed); cultural transmission producing upward parameter drift via contagion rather than biological reproduction. All patterns emerge from the eight rules. No agent was programmed to produce any of them.

Parameters: base_habitation_rate=1.015, base_status_impulse_sensitivity=0.03, cultural_selection_rate=0.005, resource_acquisition_drive=8 days, status_consumption_conversion=15 days, carrying_capacity=3000 days, fitness_signal_visibility=0.0, adaptive_recontextualisation=0.0.

Figure 4. Main simulation results: key time-series across 300 ticks.

5. Statistical and modelling assessment

5.1 What the robustness finding means

The 100% rate across 500 LHS runs is a qualitative robustness claim. The range (1.372 to 5.415 in earlier runs) confirms that parameter values affect magnitude considerably — the finding is not that all runs produce the same gap but that all runs produce a positive gap. The ABC fitting further confirms that the model structure, not just parameter choices, drives this outcome.

5.2 Sobol versus Pearson — why it matters

Pearson correlations underestimated the importance of status impulse sensitivity because they capture only linear independent effects. The Sobol analysis reveals that 83% of status impulse sensitivity's total effect ($ST=0.26$) operates through interactions with other parameters — primarily `status_consumption_conversion`. This has a direct intervention implication: the two social comparison parameters form an amplifying pair. Interventions targeting one without the other will underperform.

5.3 ABC limitations

Rejection ABC at the 5th percentile is a relatively coarse approximation. Sequential Monte Carlo ABC or population Monte Carlo ABC would provide sharper posteriors with fewer model evaluations. The weakly constrained posteriors for social comparison parameters are partly a consequence of the method's resolution — more sophisticated ABC variants combined with a richer summary statistic set (including inequality and wellbeing data alongside EOD) might constrain these parameters further.

5.4 Remaining structural limitations

- **Cultural transmission:** prestige-biased toward consumption norm only in the base model. Bidirectional transmission is implemented but the wellbeing-prestige effect is small at 50-tick timescales because cultural selection (0.5%/tick) is genuinely slow. This is consistent with the paper's argument; it is not a model failure.
- **Consumption soft ceiling:** the logistic constraint prevents the hard artefact but consumption still converges in long runs. Long-run sensitivity analysis (150+ ticks) remains limited.
- **Equifinality:** three simultaneous trajectory matches substantially reduce but do not eliminate the equifinality concern. Establishing causal necessity requires the empirical programme.

6. Testable predictions for the empirical programme

- Communities with higher ecological footprints should show larger consequence separation — measurable through cross-sectional survey using adapted Basic Psychological Needs Satisfaction scale (Deci & Ryan, 2000).
- The ABC posterior constrains habituation rate to 1.005–1.020. Communities should show neurobiological habituation in this range — testable through longitudinal measures of reward sensitivity to identical stimuli across time.
- The Sobol interaction finding predicts that interventions targeting both social comparison sensitivity and anxiety-to-consumption conversion simultaneously will produce larger effects than either alone — a falsifiable prediction for intervention trial design.
- Status comparison interventions should show faster observable effects over 5-year horizons. Habituation-targeted interventions (community practice, physical presence, mastery) should show stronger effects at 10+ years, consistent with the model’s timescale structure.
- Reproductive physiology measures should correlate with allostatic load and EDC exposure independently of deliberate fertility choices, controlling for demographic variables.

6. References

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